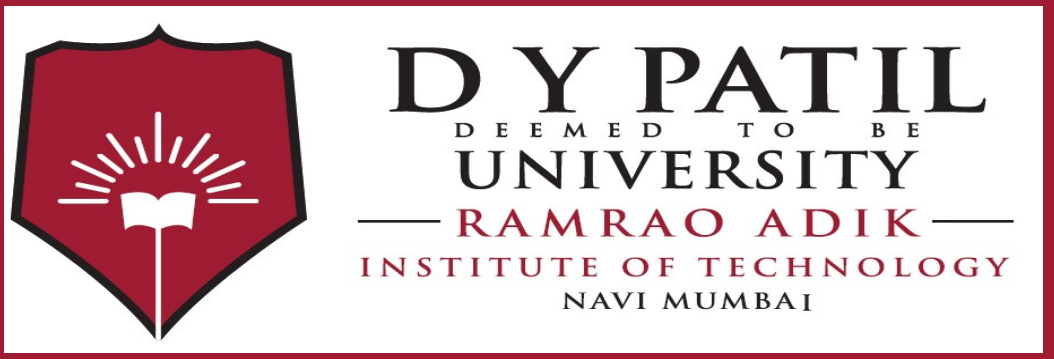


Embedded System
(Multidisciplinary Minor)
Module 6: Introduction to ARM (Advanced RISC Machine)

ARM Introduction

Founded in November 1990

- Spun out of Acorn Computers

Designs the ARM range of RISC processor cores

Licenses ARM core designs to semiconductor partners who fabricate and sell to their customers.

- ARM does not fabricate silicon itself

Also develop technologies to assist with the design-in of the ARM architecture

- Software tools, boards, debug hardware, application software, bus architectures, peripherals etc



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ARM Introduction

- Leading provider of 32-bit embedded RISC microprocessors, 75% of market
  - High performance
  - Low power consumption
  - Low system cost
- Solutions for
  - Embedded real-time systems for mass storage, automotive, industrial and networking applications
  - Secure applications - smartcards and SIMs
  - Open platforms running complex operating systems

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ARM Connected Community – 700+



Huge Range of Applications



Equipment Adopting 32-bit ARM Microcontrollers

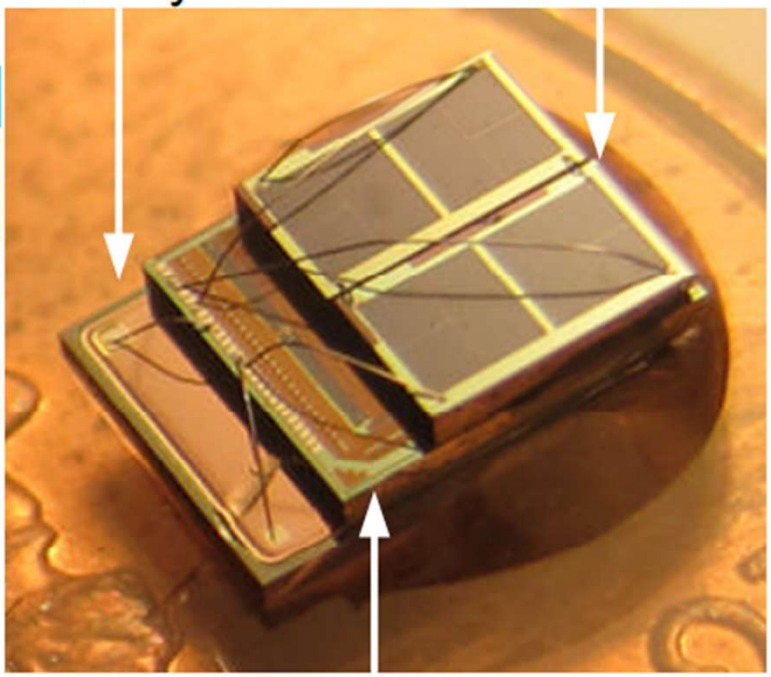


World's Smallest ARM Computer?

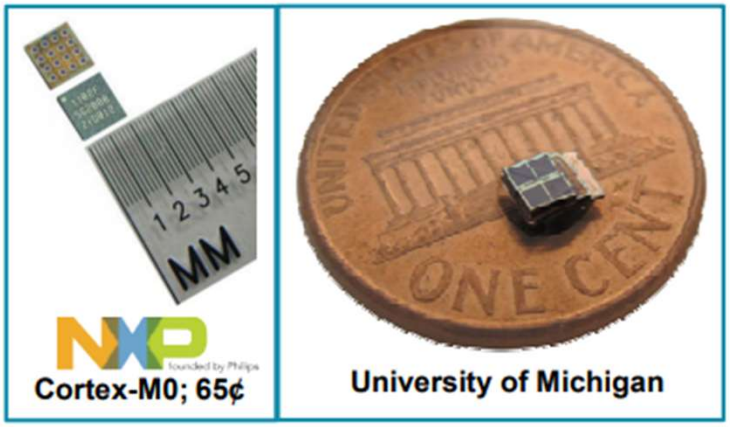


Wireless Sensor Network

Sensors, timers
Cortex-M0 +16KB RAM 65nm UWB Radio antenna
10 kB Storage memory ~3fW/bit
12µAh Li-ion Battery



Processor, SRAM and PMU



Cortex-M0; 65¢

University of Michigan

Wirelessly networked into large scale sensor arrays

World's Largest ARM Computer?



4200 ARM powered Neutrino Detectors

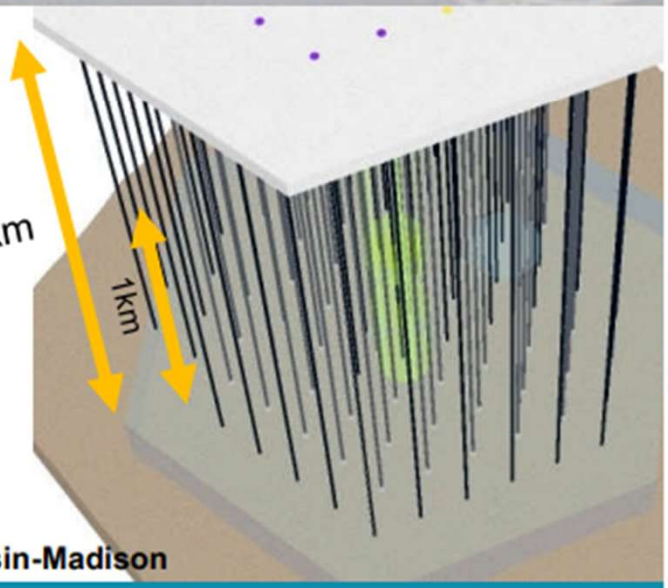




70 bore holes 2.5km deep

60 detectors per string starting 1.5km down

1km^3 of active telescope



Work supported by the National Science Foundation and University of Wisconsin-Madison

2/3

- ARMv1

  First version of ARM processor

  26-bit addressing, no multiply / coprocessor

- ARMv2

  ARM2, First commercial chip

  Included 32-bit result multiply instructions / coprocessor support

3/3

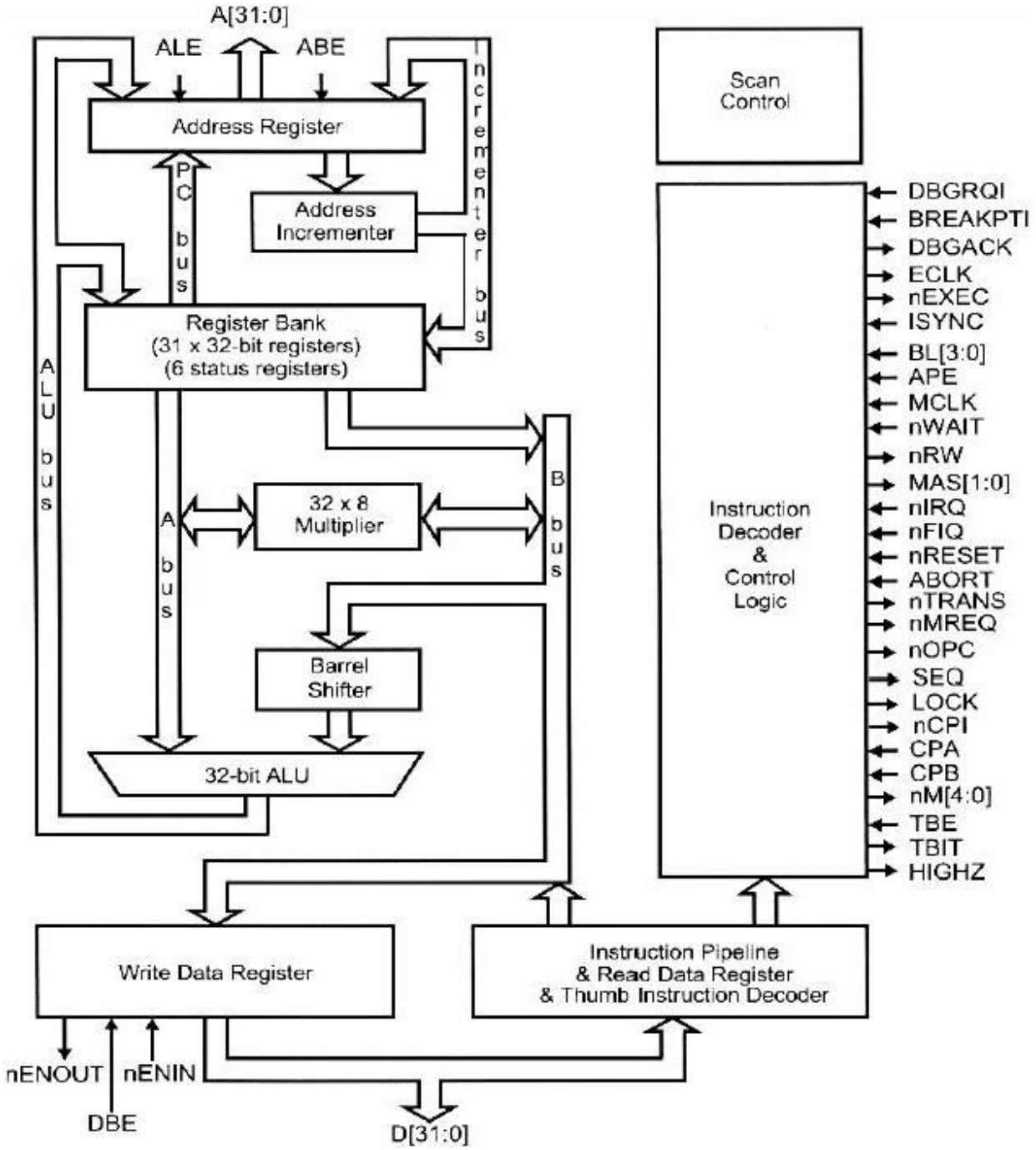
- ARMv2a
  - ARM3 chip with on-chip cache
  - Added load and store
  - cache management

- ARMv3
  - ARM6, 32 bit addressing, virtual
    memory support

ARM Processor Core

- Current low-end ARM core for applications like digital mobile phones
- TDMI
  - **T**: Thumb, 16-bit instruction set
  - **D**: on-chip Debug support, enabling the processor to halt in response to a debug request
  - **M**: enhanced Multiplier, yield a full 64-bit result, high performance
  - **I**: EmbeddedICE hardware
- Von Neumann architecture
- 3-stage pipeline

ARM Core Diagram



The Registers

- ARM has 37 registers all of which are 32-bits long.
  - 1 dedicated program counter
  - 1 dedicated current program status register
  - 5 dedicated saved program status registers
  - 30 general purpose registers
- The current processor mode governs which of several banks is accessible.

  Each mode can access
  - a particular set of **r0-r12** registers
  - a particular r13 (the stack pointer, sp) and r14 (the link register)
  - the program counter, r15 (pc)
  - the current program status register, cpsr

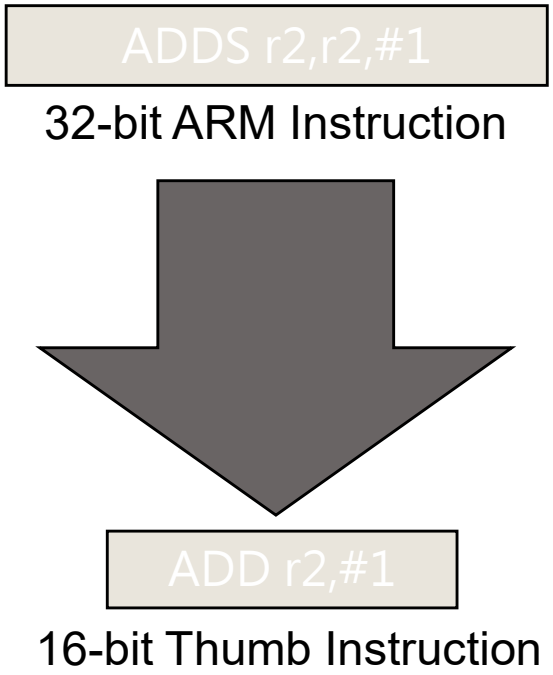
  Privileged modes (except System) can also access
  - a particular spsr (saved program status register)

DIFFERENT STATES

- When the processor is executing in ARM state:
  - All instructions are 32 bits wide
  - All instructions must be word aligned
- When the processor is executing in Thumb state:
  - All instructions are 16 bits wide
  - All instructions must be halfword aligned
- When the processor is executing in Jazelle state:
  - All instructions are 8 bits wide
  - Processor performs a word access to read 4 instructions at once

Thumb

- Thumb is a 16-bit instruction set
  - Optimised for code density from C code (~65% of ARM code size)
  - Improved performance from narrow memory
  - Subset of the functionality of the ARM instruction set
- Core has additional execution state - Thumb
  - Switch between ARM and Thumb using `BX` instruction



For most instructions generated by compiler:

- Conditional execution is not used
- Source and destination registers identical
- Only Low registers used
- Constants are of limited size
- Inline barrel shifter not used

CPSR in ARM

- **Current Program Status Register | CPSR**
- CPSR in ARM is used in core to monitor and control internal functions. The vacant area has been set aside for future development.
- Flags, status, extension, and control are the four fields of the CPSR, each of which is 8 bits wide.
- Status and extension fields are kept for future use in current designs.
- Several ARM processor cores include extra bits allocated to the J bit (available only on Jazelle enabled processing which executes 8-bit instructions).

CPSR

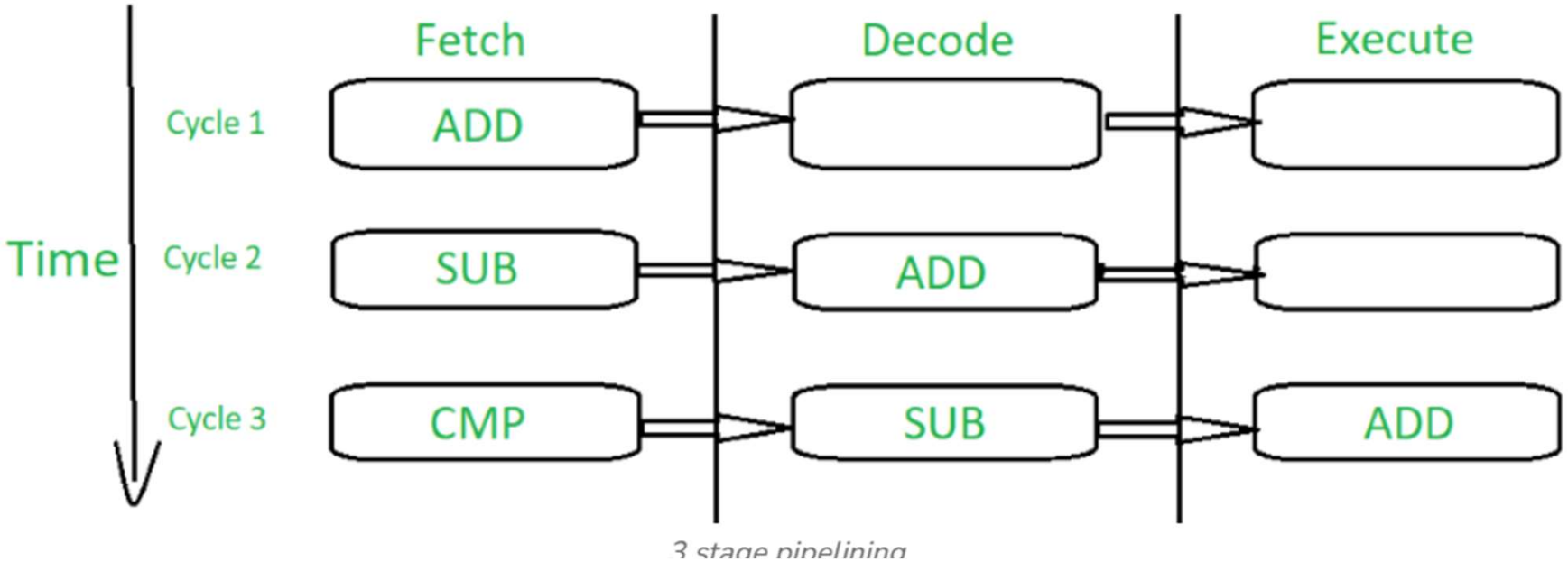


Bits	Field	Description
31	N (Negative flag)	Set if result of operation is negative
30	Z (Zero flag)	Set if result of operation is zero
29	C (Carry flag)	Set if operation produces a carry or borrow
28	V (Overflow flag)	Set if arithmetic overflow occurs
27	Q (Saturation flag)	Indicates saturation in DSP instructions
26–25	Reserved	(Not used in most cases)
24	J (Jazelle bit)	Used for Java bytecode execution (optional)
23–20	Reserved	—
19–8	GE bits (for SIMD)	Greater-than-or-Equal flags (optional)
7	E (Endian bit)	0 = Little-endian, 1 = Big-endian
6	A (Abort disable)	1 = disables imprecise data aborts
5	I (IRQ disable)	1 = disables IRQ interrupts
4	F (FIQ disable)	1 = disables FIQ interrupts
3	T (Thumb state)	0 = ARM mode (32-bit), 1 = Thumb mode (16-bit)
2–0	M[4:0] (Mode bits)	Selects current processor mode

Pipelining in ARM

- A **Pipelining** is the mechanism used by **RISC(Reduced instruction set computer)** processors to execute instructions,
- by speeding up the execution by **fetching** the instruction, while other instructions are being **decoded and executed** simultaneously.
- Which in turn allows the memory system and processor to work continuously.
- The pipeline design for each ARM family is different.

3 Stages Pipelining in ARM

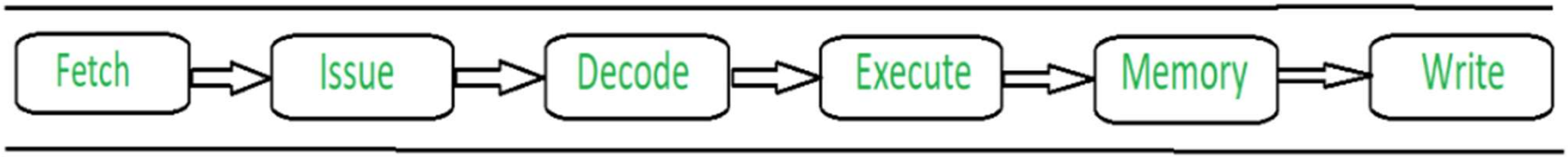


ARM 7- 3 Stage Pipelining

ARM 9- 5 Stage Pipelining ARM10- 6Stage Pipelining



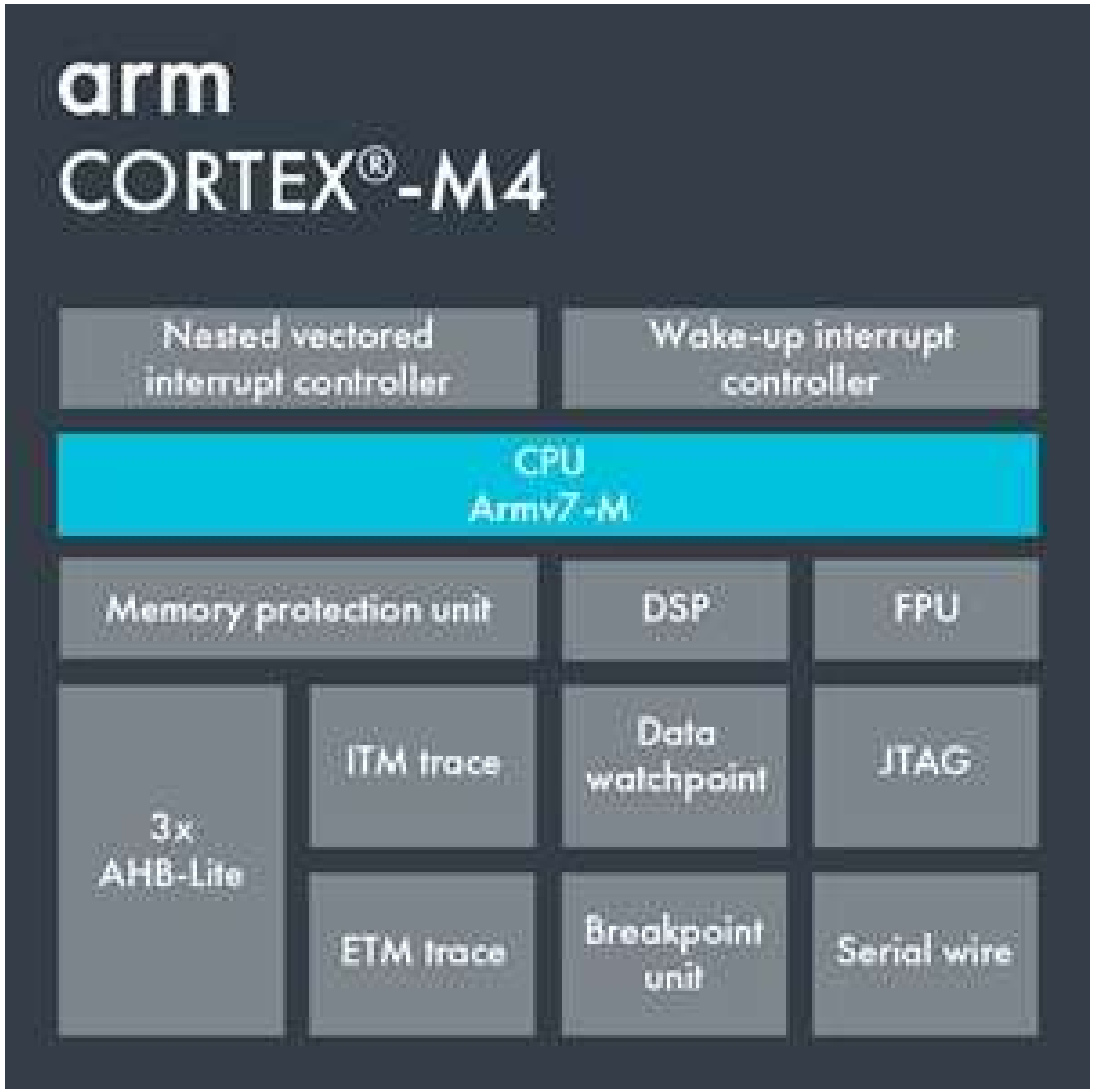
6 Stage Pipelining –ARM10



ARM CORTEX –M4 Processor

- The Arm Cortex-M4 processor is a highly-efficient embedded processor.
- The Cortex-M4 processor is developed to address digital signal control markets that demand an efficient, easy-to-use blend of control and signal processing capabilities.
- The combination of high-efficiency signal processing functionality with the low-power, low cost and ease-of-use benefits of the Cortex-M family of processors satisfies many markets. These industries include motor control, automotive, power management, embedded audio and industrial automation markets.

ARM Cortex Architecture



Resources

- https://developer.arm.com/documentation/ddi0406/c/Appendices/ARMv6-Differences/System-level-register-support/Program-Status-Registers--PSRs-?lang=en
- https://developer.arm.com/Processors/Cortex-M4

Thank You

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